

Complete Corrected Python Notes: Syntax and Structure, First Program, Variables and Constants

1. Python Syntax and Structure

Python syntax refers to the rules used for writing Python programs. Python is designed to be simple and readable.

Main Features of Python Syntax

a) Indentation

- Python uses indentation (spaces or tabs) to define blocks of code.
- Usually, 4 spaces are used.
- Incorrect indentation causes errors.

Example:

```
if 10 > 5:  
    print("10 is greater")
```

b) Statements

A statement is an instruction given to the computer.

Examples:

```
x = 5  
y = 10  
print(x+y)
```

c) Comments

Comments explain code and are ignored by Python.

Single-line comment:

```
# This is a comment
```

Multi-line comment:

```
"""This is a multi-line comment"""
```

d) Case Sensitivity

Python treats name and Name as different variables.

e) Keywords

Examples: if, else, while, for, class, break, return, import

Structure of a Python Program:

```
# Comments
```

```
name = "Python"
```

```
print("Welcome to", name)
print("Learning Python")
```

2. First Python Program

The first program traditionally written by beginners is Hello World.

Program:

```
print("Hello, World!")
```

Output:

Hello, World!

Explanation:

- print() displays output on the screen.
- Text inside quotation marks is called a string.

More Example:

```
print("My first Python program")
print("I am learning Python")
```

3. Variables in Python

Definition:

A variable is a name that refers to a value stored in memory. It acts as a reference to data so it can be used and modified during program execution.

Syntax:

```
variable_name = value
```

Example:

```
name = "John"
age = 20
marks = 90.5
```

```
print(name)
print(age)
print(marks)
```

Rules for Naming Variables:

1. Can contain letters, numbers and underscores.
2. Cannot start with a number.
3. Cannot contain spaces.
4. Cannot use keywords.

Valid:

```
student_name = "Alex"  
marks1 = 90  
_age = 21
```

Invalid:

```
1name = "Alex"  
student name = "Alex"  
class = 20
```

Types of Variables:

Integer: num = 100

Float: price = 25.50

String: name = "Python"

Boolean: status = True

Multiple Assignment:

```
x,y,z=10,20,30
```

4. Constants in Python

Constants are values that should not change during program execution.

Python does not have a constant keyword.

By convention constants are written in uppercase letters.

Example:

```
PI = 3.14159
```

```
GRAVITY = 9.8
```

Difference Between Variables and Constants:

Variables:

- Values can change
- Usually lowercase

Constants:

- Values should not change
- Usually uppercase

Summary

Python syntax uses readable rules such as indentation and statements. A variable is a name that refers to a value stored in memory, while constants are used for fixed values that should remain unchanged.